

ANIMESH DHIMAN

Una, Himachal Pradesh, India | +91 94181 79744 | animeshdhiman10@gmail.com | LinkedIn | GitHub

Professional Summary

Computer Science undergraduate at Thapar Institute of Engineering and Technology (2023–2027) with hands-on experience building data-driven systems end-to-end — from Python/SQL data pipelines and predictive models to full-stack GenAI-powered applications. Co-authored a deep learning paper on infrastructure monitoring, currently under review at Elsevier. Selected participant, Amazon ML Summer School 2026. Comfortable converting ambiguous problems into data problems, prototyping quickly, and shipping working systems with strong technical ownership.

Education

Thapar Institute of Engineering and Technology

2023 – 2027

Bachelor of Engineering in Computer Science, Patiala, Punjab

Research Experience

Research Intern – Thapar Institute of Engineering and Technology

Jun 2024 – Jul 2024

- Benchmarked custom DCNN, VGG19-UNet, and ResNet50-UNet architectures for binary crack classification on a 40,000-image dataset; ResNet50 achieved 99.9% validation accuracy.
- Designed an augmentation pipeline using CLAHE, RandomResizedCrop, Otsu Thresholding, and Color-Jitter to improve model generalization on the crack-classification dataset.
- Integrated the best-performing model into a real-time YOLOv7 detection pipeline (NVIDIA RTX 2080 Ti), achieving 97.7% mAP over 100+ training epochs.
- Automated crack-width measurement and alerting via OpenCV contour analysis, aligned with IRC:SP:83 infrastructure inspection standards.
- Co-authored a paper submitted to Elsevier on deep learning-based infrastructure monitoring and empirical ML evaluation (under review, Nov 2024).

Projects

Solar Panel Efficiency Predictor

Python • Ensemble ML • XGBoost • LightGBM • CatBoost

- Built a regression pipeline in Python to predict solar panel efficiency from environmental sensor data, applying feature engineering to improve signal quality and reduce overfitting.
- Achieved 89.89% R^2 using an ensemble stacking approach combining XGBoost, LightGBM, and CatBoost.
- Automated the hyperparameter search process, reducing tuning time by over 60%.

AI-Driven Resume Screening System

Python • FastAPI • Transformers • spaCy • PostgreSQL

- Built an NLP pipeline combining HuggingFace transformer embeddings with spaCy named-entity recognition (NER) to extract skills, experience, and project details from unstructured resume text.
- Designed a custom scoring algorithm that reduced manual resume-screening time by over 70%.
- Developed scalable FastAPI REST APIs to support ATS (Applicant Tracking System) integration.
- Architected a PostgreSQL-backed data layer with optimized indexes and CTEs (Common Table Expressions) for efficient querying.
- Structured the system as a stateless, modular architecture, enabling independent deployability and horizontal scalability.

JARVIS – Local-First AI Desktop Assistant

Electron • React • TypeScript • FastAPI • Vosk • Coqui XTTS v2 • ChromaDB

- Designed a three-layer system architecture: an Electron (React + TypeScript) frontend, a Python FastAPI backend, and a local AI inference layer, communicating via WebSockets and a JWT-authenticated IPC bridge with renderer context isolation.
- Integrated Vosk offline streaming ASR (sub-200ms latency) with a WebRTC VAD noise gate and OpenWakeWord wake-word detection for hands-free interaction.
- Implemented Coqui XTTS v2 text-to-speech with voice cloning from 3-second audio samples, achieving under 300ms synthesis latency.
- Built a ChromaDB-backed vector store for semantic memory, using top-K cosine similarity search over conversation embeddings to support contextual recall and GenAI-powered natural-language querying.
- Hardened system security using Fernet (AES-128-CBC + HMAC-SHA256) encryption, OS-level keyring integration, and per-session JWT token rotation.

Technical Skills

- **Programming:** Python, C++, C, SQL, TypeScript
- **Machine Learning:** PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, LightGBM, CatBoost, Optuna, SHAP
- **Deep Learning:** CNNs, ResNet50, YOLOv7, VGG19, UNet, OpenCV
- **GenAI / NLP:** Transformers (HuggingFace), spaCy, NLTK, ChromaDB, Vosk, Coqui XTTS v2
- **App Development:** React.js, FastAPI, REST APIs, WebSockets
- **Databases:** PostgreSQL, SQLite, MySQL, SQLAlchemy
- **Tools:** Git, GitHub Actions, Electron, PyInstaller, Loguru

Achievements

- Selected Participant, Amazon ML Summer School 2026
- Solved 200+ LeetCode problems spanning graphs, trees, dynamic programming, and core data structures
- Practical experience with ensemble methods (XGBoost, LightGBM, CatBoost), Bayesian hyperparameter tuning with Optuna, and cross-validation strategies